

2.8] (9) $f_9(t) = \underbrace{\sqrt{t^4 + t^2 + 1}}_{(a)} - t \underbrace{\sqrt[3]{t^3 + t}}_{(b)}$

• $\sqrt{x+1} \Rightarrow x \geq -1 \quad x+1 \geq 0$

• (a) $t^4 + t^2 + 1 \quad \checkmark \quad \sqrt[3]{-1} = -1$

(b) $t^3 + t = t(t^2 + 1) \quad \checkmark \quad \text{as well} \quad \text{okay on all of } \mathbb{R}$

(6) $(1-t)^{-1} t^{-1/2} \mathbb{I}_{[0,1]}(t)$

$\mathbb{I}_{[0,1]}(t) = \begin{cases} 1 & t \in [0,1] \\ 0 & t \notin [0,1] \end{cases}$

(7) $\frac{\ln(t)}{1+t^2} \quad \cancel{\mathbb{I}_{[0,+\infty)}(t)}, \quad \frac{1}{1+t^2} \quad 1+t^2 \neq 0 \quad \checkmark$

* $\ln(t)$: limit as $t \rightarrow 0$ *
 $\rightarrow \frac{\ln(t)}{1+t^2} \quad \ln(t): t > 0$

Back to (9): $f_9(t) = \sqrt{t^4 + t^2 + 1} - t \sqrt[3]{t^3 + t}$

* $\int_{-\infty}^{\infty} |f(t)| dt$: converges $\leftarrow$

$\rightarrow$ ex: $f_2(t) = \ln(t) \mathbb{I}_{[0,1]}(t)$

$\int_{-\infty}^{\infty} |\ln(t) \mathbb{I}_{[0,1]}(t)| dt = \int_0^1 |\ln(t)| dt$

$\int_{-\infty}^{\infty} |f_9(t)| dt \quad \left| \quad \lim_{t \rightarrow \pm\infty} f_9(t) \right|$

trick: Multiply by 1: $\left(\frac{\sqrt{t^4 + t^2 + 1} + t \sqrt[3]{t^3 + t}}{\sqrt{t^4 + t^2 + 1} + t \sqrt[3]{t^3 + t}} \right) (f_9(t))$

$= \frac{t^4 + t^2 + 1 - t^2 (t^3 + t)^{1/3}}{\sqrt{t^4 + t^2 + 1} + t \sqrt[3]{t^3 + t}} \sim t^4$

Denom: $t^2 \sqrt{1 + t^{-2} + t^{-4}} + t^2 \sqrt[3]{1 + t^{-2}} \sim t^2$

$\downarrow$
 $t(t^3 + t)^{1/3} = t(t^3(1 + t^{-2}))^{1/3}$
 $= t(t)(1 + t^{-2})^{1/3}$

$+4$

$$\sim \frac{1}{t^2} \left(\right) \sim \frac{1}{t^2}$$

$$\lim_{t \rightarrow \pm\infty} f(t) \sim \lim_{t \rightarrow \pm\infty} t^2 = +\infty$$

$$\lim_{t \rightarrow \pm\infty} t^2 = +\infty$$

$$(2.10) \textcircled{1} \int_0^1 \frac{1}{\sqrt{x(1-x)}} dx$$

$x=0$
 $x=1$

Ⓐ Look at limit points:

$x=0$: $x(1-x)$: $\lim_{x \rightarrow 0} x(1-x) = \left(\lim_{x \rightarrow 0} x \right) \left(\lim_{x \rightarrow 0} (1-x) \right)$

$\sim \frac{1}{\sqrt{x}}$: Does Converge

$x=1$: $\lim_{x \rightarrow 1} x(1-x) = \lim_{x \rightarrow 1} (x) \lim_{x \rightarrow 1} (1-x) = \lim_{x \rightarrow 1} (1-x)$

$\sim \frac{1}{\sqrt{x-1}}$ Does Converge

$\int_0^1 \frac{1}{\sqrt{x(1-x)}} dx$: $x(1-x) = \frac{1}{4} - \left(\frac{1}{2} - x\right)^2$

$\int_0^1 \frac{1}{\sqrt{\frac{1}{4} - \left(\frac{1}{2} - x\right)^2}} dx$

$\left(\frac{1}{2} - x\right)^2 = \left(x - \frac{1}{2}\right)^2$

$u = x - \frac{1}{2}$

$du = dx$

$x=0$: $u = -\frac{1}{2}$

$x=1$: $u = \frac{1}{2}$

$\int_{-1/2}^{1/2} \frac{1}{\sqrt{\frac{1}{4} - u^2}} du$

 $= \arcsin\left(\frac{u}{1/2}\right) = \arcsin(2u) \Big|_{-1/2}^{1/2}$

$= \arcsin(1) - \arcsin(-1) = \pi$

$\frac{\pi}{2} - (-\pi/2) = \pi$

$= \pi$ ✓

Ⓐ $\int_0^{\sqrt{3}/2} \frac{1}{\sqrt{3-4x^2}} dx$

$\frac{1}{\sqrt{3-4x^2}}$

$3-4x^2 \geq 0$

$\frac{3}{4} \geq x^2$

$x \leq \sqrt{\frac{3}{4}}$

$\frac{\sqrt{3}}{2}$: $\bullet 3-4x^2 = (\sqrt{3}-2x)(\sqrt{3}+2x) \bullet$

$\lim_{x \rightarrow \frac{\sqrt{3}}{2}} (\sqrt{3}-2x) = \left[\lim_{x \rightarrow \frac{\sqrt{3}}{2}} (\sqrt{3}-2x) \right] (2\sqrt{3})$

$\left(\frac{1}{\sqrt{3}} \right) \int \frac{1}{\sqrt{3-4x^2}} dx \sim \sqrt{\quad}$: also convergent

$$\int_0^{\frac{\sqrt{3}}{2}} \frac{1}{\sqrt{3-4x^2}} dx = \int_0^{\frac{\sqrt{3}}{2}} \frac{1}{2\sqrt{\frac{3}{4}-x^2}} dx = \frac{1}{2} \arcsin\left(\frac{x}{\sqrt{3}/2}\right)$$

$$\frac{1}{2} \arcsin\left(\frac{2x}{\sqrt{3}}\right) \Big|_0^{\sqrt{3}/2} = \frac{1}{2} \arcsin(1) - \frac{1}{2} \arcsin(0) = \frac{1}{2} \frac{\pi}{2} - \frac{1}{2} 0 = \frac{\pi}{4}$$

2.4] $\int_0^\infty \frac{1}{x^\alpha (1+x)^\beta} dx$ $f(x) = \frac{1}{x^\alpha (1+x)^\beta}$

Step 1: look at limit points

$x=0$: $x^\alpha (1+x)^\beta \rightarrow x^\alpha$

$$\lim_{x \rightarrow 0} x^\alpha (1+x)^\beta = \lim_{x \rightarrow 0} x^\alpha \underbrace{\lim_{x \rightarrow 0} (1+x)^\beta}_{=1}$$

$\therefore f(x) \underset{x \rightarrow 0}{\sim} \frac{1}{x^\alpha}$

$\int_0^\infty \frac{1}{x^\alpha} dx$ $\alpha < 1$
 $\alpha < 0$ $\alpha = 0$, $\alpha > 0$

$\int \frac{1}{x^{n+1}} dx = \ln|x|$ $\alpha \neq 1$

$\int \frac{1}{x^{-1}} dx = \int x dx = \frac{1}{2} x^2$ $\alpha \leq 0$

$\int \frac{1}{x^{-n}} dx = \int x^n dx \sim x^{n+1}$ $\alpha \neq 1$

$\int \frac{1}{x^2} \sim \frac{1}{x}$ $\alpha \neq 1$

$\int \frac{1}{x^3} \sim \frac{1}{x^2} \xrightarrow{x \rightarrow \infty} 0$

$\alpha < 1$ $x \rightarrow 0$

(b) $x \rightarrow \infty$: $x^\alpha (1+x)^\beta \xrightarrow{x \rightarrow \infty}$

$\lim_{x \rightarrow \infty} (1+x) \approx x$

$f(x) \xrightarrow{x \rightarrow \infty} \frac{1}{x^\alpha x^\beta} = \frac{1}{x^{\alpha+\beta}}$

$\int_0^\infty$

$\int_0^\infty$

$$\int_0^1 \frac{1}{x^{\alpha+\beta}} dx$$

$$\alpha + \beta > 1$$

$$\underline{\alpha < 1}, \quad \underline{\beta > 1 - \alpha}$$

$$2.12] \quad a > 0 : \quad \int_0^\infty e^{-ax} \ln(x) dx$$

Step 1: look at limits

$$\underline{x=0}: \quad \lim_{x \rightarrow 0} e^{-ax} \ln(x) \sim \ln(x)$$

$$\underline{x \rightarrow \infty}: \quad \lim_{x \rightarrow \infty} \left(\underbrace{e^{-ax}}_{\uparrow} \underbrace{\ln(x)}_{\sim} \right) \quad \quad \quad e^{-ax} \xrightarrow{x \rightarrow \infty} 0$$

$$\int_0^\infty e^{-ax} \ln(x) dx$$

$$\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} dx$$

